Nevin Zheng

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Summary

Software engineer specializing in database and distributed data systems. ~5 years on Google BigQuery, then ~6 months at a seed-stage startup contributing to open-source Apache Iceberg and Gravitino. Looking to own hard problems end to end on a database or data infrastructure team that cares about doing things well, and to keep growing in technical leadership.

Professional Experience

Datastrato | **Software Engineer** | Remote | Jul 2026 – Nov 2026

Seed-stage startup building on Apache Gravitino (open-source data catalog) and an enterprise Iceberg REST Catalog.

- Designed and implemented Apache Iceberg table encryption in the enterprise Iceberg REST Catalog, with keys referenced in an external KMS, key metadata served to query engines, and catalog-side metadata integrity checks.
- Cut a Gravitino CI integration-test suite from ~1 hour to under 10 minutes (~10x) by profiling container lifecycles and moving to a shared Testcontainers instance per run; merged into Apache Gravitino.
- Designed and prototyped tombstone-based soft delete and incremental / full-refresh materialized view maintenance against the draft Iceberg materialized view spec.
- Contributed to Apache Iceberg spec discussions on REST catalog read restrictions, materialized views, and table encryption.
- Self-initiated and prototyped Loom, a release-qualification CI/CD platform to catch regressions before release.
- Led cloud security incident response and recommended review-gated infrastructure-as-code.

Google | **Software Engineer, BigQuery** | Seattle, WA | 2021 – 2026

Project: Streaming for BigLake

- Owned productionization of a novel streaming feature from local prototype to customer preview, driving the end-to-end testing strategy, release process, and SRE collaboration.
- Drove design refinements and API negotiations across 3 core teams (BigQuery, WriteAPI, SRE) in a complex microservices architecture.
- Implemented fixes during a critical production outage and contributed to the postmortem.

Project: Row Level & Column Level Security for BigLake

- Owned end-to-end integration of row- and column-level security for a new table format, reaching feature parity for an enterprise product.
- Designed a validation strategy across query surfaces (BigQuery API, Read API) that caught several critical bugs in dependent systems, preventing potential data leaks.

Projects

- strata-db (Rust): SQL database engine built from scratch, covering SQL parsing, query execution, key/row encoding, schema catalog, LSM storage, WAL, and Postgres wire protocol.
- mammoth-cache (Rust): self-hosted build cache for Gradle, Maven, and Bazel.
- mori (Rust): worktree manager for running multiple coding agents in parallel.

Technical Skills

- Languages & Tools: Rust, C++, Java, Python, SQL, Shell, Terraform, Pulumi, Gradle, Bazel
- Distributed Systems: Database Internals (LSM Trees, Consensus), Query Processing & Optimization, Fault-Tolerant Architectures, Data Lakes, Streaming Data Pipelines
- Platforms: Google Cloud Platform (GCP), BigQuery, Google Spanner

Education

Carnegie Mellon University | Master of Science, Computer Engineering | 2019 – 2021

Specialization in Database Systems with Professor Andy Pavlo.

Boston University | Bachelor of Science, Computer and Electrical Engineering (Double Major) | 2013 – 2017